Figure 2.1. Architects of the Copenhagen interpretation at the Niels Bohr Institute, 1936. Left to right: Bohr, Heisenberg, and Pauli.

Yet for all the successes of wave mechanics—and for all Schrödinger's bluster—it seemed that in the areas they overlapped, Schrödinger's wave mechanics gave the same results as Heisenberg's matrix mechanics. Schrödinger's theory, like Heisenberg's, reproduced the spectrum of the hydrogen atom perfectly: the different energy levels of Bohr's atom were, in Schrödinger's theory, associated with energy "eigenstates," special wave functions with constant energies. As Schrödinger soon discovered, matrix mechanics and wave mechanics were mathematically equivalent, using different tools to describe the same ideas: a single new theory of quantum mechanics. Problems like the brightness of spectral lines had been solved first with wave mechanics only because Schrödinger's equation was mathematically easier to handle than Heisenberg's matrices in most situations. But the two versions of quantum mechanics still differed radically in their interpretation. Schrödinger was sure that he could find a way to interpret all quantum phenomena as the smooth movement of the waves his equation described. Heisenberg was unconvinced. "The more I think about the physical portion of the Schrödinger theory, the more repulsive I find it," he wrote to Pauli. "What Schrödinger writes about the visualizability of his theory is 'probably not quite right,' in other words it's crap."

Yet Schrödinger's waves seemed more natural to most physicists than Heisenberg's matrices. Heisenberg, frustrated by the situation and fearful that Schrödinger's ideas would eclipse his own, wrote to his mentor Bohr. Bohr in turn wrote to Schrödinger, inviting him to visit Copenhagen to have "some discussions for the narrower circle of those who work here at the Institute, in which we can deal more deeply with the open questions of atomic theory." Schrödinger arrived by train on the first of October, 1926, and the debate started immediately, as Heisenberg later recalled:

> Bohr's discussions with Schrödinger began at the railway station and were continued daily from early morning until late at night. Schrödinger stayed in Bohr's house so that nothing would interrupt the conversations. And although Bohr was normally most considerate and friendly in his dealings with people, he now struck me as an almost remorseless fanatic, one